

ANDEER, Switzerland

WMO: 067870

Lat: 46.6101N

Lon: 9.4319E

Elev: 988

StdP: 90.00

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.3	-10.7	-16.5	1.0	-11.0	-13.9	1.3	-9.2	8.5	5.2	7.4	5.1	0.7	120	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	28.7	17.3	26.9	16.8	24.9	16.1	18.2	26.0	17.5	24.7	16.7	23.2	3.8	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.9	12.7	20.0	15.1	12.1	19.2	14.3	11.5	18.7	55.8	26.2	53.1	25.0	50.8	23.2	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.1	6.0	5.2	-15.8	31.6	3.2	0.7	-18.1	32.2	-20.0	32.6	-21.9	33.0	-24.2	33.5		
(5)			-16.3	20.2	3.2	0.9	-18.6	20.9	-20.4	21.4	-22.2	21.9	-24.5	22.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-1.9	-0.7	3.8	8.1	11.3	15.8	17.2	17.1	13.0	9.0	4.0	-0.4		
	DBStd	7.63	3.89	4.64	3.41	3.44	3.39	3.28	3.03	2.98	2.98	3.74	4.09	4.06		
	HDD10.0	1537	370	299	192	74	26	1	0	0	8	64	181	322		
	HDD18.3	3800	628	533	449	307	217	89	56	59	162	291	429	580		
	CDD10.0	836	0	0	1	17	67	175	224	221	97	32	2	0		
	CDD18.3	57	0	0	0	0	0	13	22	22	1	0	0	0		
	CDH23.3	764	0	0	0	0	22	206	271	249	16	0	0	0		
	CDH26.7	172	0	0	0	0	2	48	64	58	0	0	0	0		
(14) Wind	WSAvg	1.4	1.1	1.3	1.6	1.9	1.9	1.6	1.5	1.3	1.2	1.3	1.2	1.1		
(15) Precipitation	PrecAvg	1208	57	54	60	86	115	136	135	147	120	112	125	63		
	PrecMax	1686	117	167	144	218	302	268	220	237	274	348	512	132		
	PrecMin	808	11	16	8	16	42	40	65	20	30	9	27	2		
	PrecStd	219	34	39	34	49	62	67	43	46	77	70	105	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.0	11.4	17.3	21.9	26.2	30.5	30.9	30.7	25.2	21.2	15.5	10.8		
		MCWB	4.3	4.5	7.6	11.4	14.8	17.5	18.0	17.5	15.5	12.6	8.8	5.7		
	2%	DB	7.1	8.5	14.6	19.3	23.1	28.0	28.5	28.2	23.2	19.0	13.3	9.1		
		MCWB	3.1	3.3	6.7	10.6	13.4	16.9	17.4	17.3	14.9	11.8	7.9	4.7		
	5%	DB	5.5	6.9	12.3	17.1	20.8	25.7	26.5	26.3	20.9	17.1	12.0	7.6		
		MCWB	2.2	2.6	5.9	9.2	12.5	16.1	16.8	16.8	13.9	10.8	7.3	3.7		
	10%	DB	3.7	5.4	10.0	15.2	18.6	23.4	24.3	24.0	18.8	15.2	10.5	5.8		
		MCWB	1.4	1.7	5.0	8.3	11.5	15.1	16.2	16.2	13.3	10.3	6.6	2.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.0	5.5	8.3	11.9	15.4	19.1	19.2	19.4	16.6	13.7	9.6	6.6		
		MCDB	7.9	9.6	15.2	20.1	24.1	28.2	27.3	25.8	22.5	19.1	13.7	9.3		
	2%	WB	3.7	3.8	7.2	10.8	14.2	17.6	18.2	18.2	15.6	12.5	8.4	5.3		
		MCDB	6.3	7.2	13.4	18.4	21.8	25.4	26.2	25.5	21.3	16.8	12.3	8.6		
	5%	WB	2.7	2.9	6.2	9.7	12.9	16.6	17.5	17.4	14.7	11.7	7.6	4.1		
		MCDB	4.8	6.2	11.3	15.9	19.2	24.2	24.7	24.3	19.3	15.3	11.2	6.9		
	10%	WB	1.6	2.1	5.2	8.6	11.9	15.7	16.7	16.6	13.9	10.9	6.9	2.9		
		MCDB	3.3	5.0	9.4	14.2	17.6	22.1	22.8	22.7	17.7	14.4	10.1	5.4		
(35) Mean Daily Temperature Range	MDBR		7.7	8.9	10.3	11.2	10.9	11.6	11.8	11.9	10.7	10.1	8.2	7.5		
	5% DB	MCDBR	9.4	11.0	14.7	15.0	15.4	16.4	16.1	15.9	14.6	13.7	10.8	9.9		
		MCWBR	6.1	6.6	7.9	7.5	7.2	6.9	6.6	6.7	7.2	7.3	6.1	6.0		
	5% WB	MCDBR	8.0	10.1	13.0	13.9	13.9	14.6	13.9	13.8	12.3	11.3	10.0	8.5		
		MCWBR	5.4	6.3	7.2	7.2	6.6	6.3	6.4	6.4	6.4	6.5	5.7	5.4		
(40) Clear-Sky Solar Irradiance	taub		0.256	0.283	0.330	0.381	0.397	0.407	0.402	0.394	0.366	0.335	0.289	0.253		
	taud		2.316	2.242	2.172	2.146	2.230	2.257	2.276	2.318	2.374	2.433	2.432	2.356		
	Ebn at Noon		849	888	888	871	868	859	858	851	844	821	802	811		
	Edh at Noon		74	99	125	141	135	132	128	118	102	83	67	64		
(44) All-Sky Solar Radiation	RadAvg		1.56	2.52	3.73	4.45	4.79	5.63	5.60	4.86	3.78	2.50	1.53	1.30		
	RadStd		0.19	0.30	0.54	0.72	0.48	0.50	0.49	0.44	0.37	0.29	0.23	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (117 neighbors)		N/A	N/A	N/A	+0.70	+0.40	+0.32	N/A	N/A	+59	+16

Nomenclature: See separate page